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RESEARCH

Reliable cross-view neighborhood relation transfer and high-order semantic structure mining for incomplete multi-view partial multi-label classification

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Abstract

Multi-view multi-label classification has attracted increasing attention because it can characterize complex real-world data from multiple perspectives. However, the simultaneous presence of missing views and labels often causes incomplete feature information and uncertain data distributions, hindering the extraction of discriminative semantics. To address this problem, we propose a unified framework termed **Re**liable **C**ross-View Neighborhood Relation Transfer and **H**igh-Order **S**emantic Structure **M**ining (ReCHSM). Specifically, we design a reliable cross-view completion mechanism that transfers neighborhood relations from observed views to reconstruct missing ones. To reduce erroneous transfer caused by cross-view heterogeneity, the mechanism evaluates candidate-neighbor reliability and employs a lightweight nonlinear network to refine the reconstructed features. To further extract structural information from the completed representations, we introduce a random-walk-based strategy that propagates the repaired local neighborhoods to capture high-order semantic dependencies. Furthermore, considering that individual views provide different discriminative information and that the quality of completed views may vary, we develop a label-semantic-consistency-driven module to assess view quality and adaptively assign fusion weights. Extensive experiments on five public datasets demonstrate the effectiveness and competitive performance of ReCHSM.

Keywords Incomplete multi-view partial multi-label classification · View completion · High-order semantic structure mining · View quality assessment

1 Introduction

As modern data collection technologies advance, samples are increasingly represented by multiple complementary views. For example, a web page can be characterized by its page content and anchor text (Zhao et al. 2017), while an individual can be described by face, fingerprint, signature, and iris information (He et al. 2023). Multi-view learning integrates such complementary information to improve representation quality (Fang et al. 2022; Hu et al. 2022; Zhu et al. 2022, 2020). Meanwhile, multi-label classification (MLC) assigns multiple semantic labels to each sample; for instance, a natural scene image may contain "sky", "tree", and "cloud", whereas a document may involve "politics", "economy", and "sports" (Liu et al. 2022). Combining these two paradigms gives rise to Multi-View Multi-Label Classification (MvMLC), which has attracted increasing attention.

Existing MvMLC methods exploit cross-view complementarity and label dependencies in different ways. LCBM (Sun and Zong 2021) models label dependencies through a conditional Bernoulli mixture in a view-shared latent subspace and employs a Gaussian Mixture Variational Autoencoder for approximate inference. LSA-MML (Zhang et al.

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2018) instead learns a shared representation through matrix factorization and aligns view-specific latent bases in kernel spaces via HSIC. Recent studies in neighboring multimodal and multi-view fields have also emphasized heterogeneous representation alignment, adaptive information fusion, and relational dependency modeling (Chang et al. 2026; Jiang et al. 2025, 2026a; Zheng et al. 2026). These advances suggest that effective multi-view learning should jointly account for cross-view discrepancies, unequal view contributions, and complex structural relationships.

However, most existing MvMLC methods assume that all views and labels are fully available. In real-world applications, complete multi-view observations and label supervision are often difficult to obtain (Liu et al. 2024; Zeng et al. 2024). Sensor failures, viewpoint occlusion, transmission anomalies, and data corruption may cause missing views (Li et al. 2024; Zhao et al. 2017). Meanwhile, uncertainty and human error in manual annotation may result in missing labels (Li et al. 2024). These dual forms of incompleteness motivate Incomplete Multi-view Partial Multi-label Classification (IMvPMLC).

Existing IMvPMLC methods address dual incompleteness from different perspectives. Early approaches, including iMvWL (Tan et al. 2018) and NAIM3L (Li and Chen 2022), mainly rely on shared-subspace learning, missing indicators, and label-structure regularization. More recent deep models employ instance-level contrastive alignment in DICNet (Liu et al. 2023c), Transformer-based view–category interactions in LMVCAT (Liu et al. 2023d), multi-scale locality preservation in MSLPP (Long et al. 2024), label-driven graph learning and sample-level quality discrimination in RANK (Liu et al. 2025), and dual-space structural constraints with uncertainty-aware fusion in DSPNF (Jiang et al. 2026b). Although effective, these methods primarily suppress, bypass, or compensate for unavailable views during representation learning and fusion rather than explicitly repairing their local neighborhood structures.

Recovery-driven methods provide another solution by inferring unavailable views. LOGIC (Zheng et al. 2025) generates missing views from a common latent representation under an information-theoretic framework, whereas relation-transfer methods infer target views using neighborhood relations preserved in observable source views (Wang et al. 2023). However, latent generation may not explicitly preserve target-view local geometry, and direct relation transfer may introduce unreliable candidates when heterogeneous views provide inconsistent neighborhood evidence. In addition, graph constraints constructed from incomplete or coarsely recovered representations may propagate distorted local relations into high-order dependencies.

To address these limitations, we propose Reliable Cross-View Neighborhood Relation Transfer and High-Order Semantic Structure Mining (ReCHSM) for classification with simultaneously missing views and labels. ReCHSM follows a reliability-aware completion-to-structure learning process. It first reconstructs each missing view through cross-view neighborhood relation transfer, reliability-based candidate selection, weighted aggregation, and nonlinear refinement. It then constructs relation graphs from the completed representations and applies multi-step random-walk propagation to capture high-order semantic dependencies. Finally, an auxiliary label-semantic-consistency-based mechanism estimates sample-level view quality and adaptively reduces the contribution of semantically inconsistent views during fusion.

Overall, the main contributions are summarized as follows.

- We develop a reliability-aware cross-view completion mechanism that combines cross-source candidate occurrence frequency and accumulated geometric distance for neighbor selection and weighted reconstruction. A lightweight nonlinear network further refines the reconstructed features.
- We propose a completion-guided high-order structure learning strategy that constructs relation graphs from completed view representations and extends repaired local neighborhoods through multi-step random-walk propagation.
- We introduce a label-semantic-consistency-guided sample-level fusion mechanism that emphasizes views whose pairwise relations better preserve the semantics characterized by the observed labels. Extensive experiments on five benchmark datasets further demonstrate the effectiveness and competitiveness.

2 Related work

2.1 Multi-view multi-label classification

MvMLC, which integrates multi-view learning with multi-label classification, faces the dual challenge of handling heterogeneous representations derived from diverse views while simultaneously managing multiple correlated semantic labels. To address these challenges, researchers have developed various MvMLC approaches. Among them, Liu et al. proposed lrMMC (Liu et al. 2015), which embeds a matrix completion framework with multi-view learning capabilities. It operates by first projecting all available views into a low-rank space to extract a unified low-dimensional representation. Then, it uses learned view-combination weights to capture the complementarity among different views. Zhu et al. further proposed MVLE (Zhu et al. 2018). It learns predictive latent spaces by maximizing the correlations among feature spaces, latent spaces, and the label space. It also introduces the Hilbert–Schmidt Independence Criterion (HSIC)

to balance diversity and consistency across views. Huang et al. further proposed VLSF (Huang et al. 2019). It considers unequal contributions of different views and label-specific feature associations by learning view-label-specific representations together with view contribution weights. With the continuous advancement of deep learning, Zhao et al. developed CDMM (Zhao et al. 2021b). This method utilizes a unified neural framework to integrate both multi-view consistency and diversity, achieving this by constructing an independent classifier for each view. It encourages consistent label prediction across views and further introduces HSIC-based diversity modeling, label-correlation factors, and view-contribution factors. To handle non-aligned scenarios, Zhao et al. further proposed LVSL (Zhao et al. 2023), which explicitly learns view-specific labels and combines feature-space Laplacian graph regularization with low-rank label correlation modeling to preserve the original geometric structure and better capture label dependencies. Liu et al. subsequently proposed ELSMML (Liu et al. 2023a). This approach incorporates multi-view learning into a broader framework featuring dimensionality reduction and manifold regularization, all for the purpose of identifying high-level latent semantic information within the labels. Beyond task-specific MvMLC research, recent studies in neighboring multimodal and multi-view fields have further investigated representation alignment, adaptive fusion, and relational dependency modeling. Jiang et al. employed modality-aligned preprocessing to reduce the distribution discrepancy between heterogeneous modalities before representation learning (Jiang et al. 2025). Chang et al. retained relatively independent modality-specific learning branches and performed adaptive late fusion to reduce interference caused by direct heterogeneous feature aggregation (Chang et al. 2026). From the perspective of relational modeling, Jiang et al. combined knowledge graphs with graph attention networks to capture explicit semantic dependencies among multimodal entities (Jiang et al. 2026a). Furthermore, Zheng et al. proposed STANCE, which tensorizes view-specific self-representation matrices to capture high-order inter-view correlations and employs sample-level attention for adaptive multi-view fusion (Zheng et al. 2026). These studies demonstrate the importance of handling cross-view heterogeneity, unequal view contributions, and complex structural dependencies. Nevertheless, they mainly focus on complete multimodal data or unsupervised multi-view clustering and do not directly address the simultaneous absence of views and semantic labels considered in IMvPMLC.

2.2 Incomplete multi-view multi-label classification

In MvMLC, the simultaneous missingness of views and labels reduces the semantic completeness of multi-view data. It increases distribution uncertainty and results in sparse supervision. As a result, feature learning and classification become much more challenging. To alleviate this issue, Zhao et al. proposed TM3L (Zhao et al. 2021a), which adopts a two-stage learning strategy to jointly model multi-view features in a shared low-dimensional subspace and further incorporates label matrix completion to mitigate the adverse effects caused by dual incompleteness. Deng et al. further proposed MVIML-TAR (Deng et al. 2026), which introduces Tucker decomposition for shared latent label learning so as to characterize cross-view consistency and latent semantic associations in incomplete multi-view data. With the advancement of deep learning, representation-learning frameworks have achieved considerable progress under incomplete settings. DICNet (Liu et al. 2023c) employs missing-aware instance-level contrastive learning to align available representations across views and uses missing indicators to exclude invalid instances during reconstruction, contrastive learning, and fusion. MTD (Liu et al. 2023b) decomposes view-level representations into shared and view-specific components and combines cross-channel contrastive learning with random fragment masking. MSLPP (Long et al. 2024) preserves multi-scale locality structures in both the original and latent spaces while reducing the influence of unavailable views through a missing-view masking strategy. RANK (Liu et al. 2025) further introduces label-driven graph embedding and a sample-level quality discriminator supervised by view-specific classification performance. DSPNF (Jiang et al. 2026b) jointly imposes label-space and random-walk-based high-order structural constraints and adopts prediction-performance- and uncertainty-driven quality supervision for adaptive fusion. More recently, SCECD-Net (Jiang et al. 2025) enhances the consistency between label semantics and view representations while modeling sample-level view contributions to reduce the interference of low-quality views during prediction. Another line of research explicitly reconstructs or generates missing-view information rather than merely shielding unavailable observations. For example, Zheng et al. proposed LOGIC (Zheng et al. 2025), which unifies complete and incomplete multi-view clustering through an information-theoretic generative model. It interprets missing-view recovery according to the principles of comprehensiveness, consensus, and compressibility and generates missing views from a common latent representation. Cross-view relation-transfer methods provide a different recovery route by transferring neighborhood relations available in source views to missing target views (Wang et al. 2023). However, latent-generation-based approaches do not explicitly guarantee the preservation of target-view local geometric structures, whereas direct relation transfer may introduce unreliable neighbors when heterogeneous source views provide inconsistent neighborhood evidence. Overall, existing IMvPMLC methods mainly mitigate missing-view effects through masking, structural regularization, or quality-aware fusion, while recovery-driven approaches reconstruct

missing information through latent generation or direct relation transfer. However, the former do not explicitly repair the local topology destroyed by missing target views, whereas the latter may introduce unreliable relations under cross-view heterogeneity. ReCHSM therefore focuses on reliability-aware topology reconstruction and uses the repaired representations as the basis for subsequent high-order semantic propagation, with label-semantic-consistency-based weighting serving as an auxiliary fusion mechanism.

3 The proposed method

This section first formulates the IMvPMLC task, then presents the overall framework of ReCHSM, and finally describes each module in detail.

3.1 Problem formulation

Let $\mathcal{X} = \{X^{(v)} \in \mathbb{R}^{n\times d_v}\}_{v=1}^{m}$ denote the original multi-view dataset with n samples and m views, where $X^{(v)}$ and d_v are the feature matrix and dimension for the v-th view. Given a label matrix $Y \in \{0,1\}^{n\times \ell}$ for ℓ categories, we set $Y_{i,j} = 1$ if sample i belongs to category j, and 0 otherwise. To model missing views and labels, we introduce indicator matrices $W \in \{0,1\}^{n\times m}$ and $G \in \{0,1\}^{n\times \ell}$. Here, $W_{i,v} = 1$ indicates that the v-th view of sample i is available, whereas $G_{i,j} = 1$ indicates that its j-th label is observed. Under this configuration, we aim to develop a classification model to accurately predict the complete label set for each sample. The primary notations are summarized in Table 1. For notational simplicity, the formulations use n samples. In implementation, pairwise relation matrices are constructed within each mini-batch and have size $b \times b$, where b is the mini-batch size.

3.2 Overall framework

Figure 1 illustrates the overall framework of ReCHSM. Given incomplete multi-view data, the reliable cross-view completion module first reconstructs the missing views and produces the completed features $\{\hat{X}^{(v)}\}_{v=1}^{m}$. Each view-specific encoder $\mathrm{En}^{(v)}$ then maps its completed feature matrix into a unified latent space, yielding the embeddings $\{Z^{(v)}\}_{v=1}^{m}$. Based on these embeddings, ReCHSM proceeds through two parallel branches. The high-order semantic structure mining branch extends the repaired local relations into multi-hop semantic dependencies and guides representation learning through $\mathcal{L}_{\text{struct}}$. In parallel, the label-semantic-consistency-based view quality assessment branch estimates sample-specific view weights B under the supervision of

Table 1 Main notations

Notation	Description	Notation	Description
n, m, ℓ	Numbers of samples, views, and categories.	k	Number of nearest neighbors.
$X^{(v)}, d_v$	Original data matrix of the v-th view ($\mathbb{R}^{n\times d_v}$) and its feature dimension d_v.	$\hat{X}^{(v)}$	Completed data matrix of the v-th view ($\mathbb{R}^{n\times d_v}$).
Y	Ground-truth label matrix ($\{0,1\}^{n\times\ell}$).	W, G	Missing-view and missing-label indicator matrices, respectively ($W \in \{0,1\}^{n\times m}$, $G \in \{0,1\}^{n\times\ell}$).
$\mathcal{I}_v$	Index set of samples whose v-th view is observed.	$D^{(v)}$	Euclidean distance matrix of the v-th view ($\mathbb{R}^{n\times n}$).
$\mathcal{V}_{\text{obs}}^{(i)}$	Set of observable source views for sample i.	$\mathcal{P}_{i,u}$	Candidate neighbor set of sample i retrieved from source view u.
$\widetilde{\mathcal{P}}_{i,u}^{(v)}$	Transferred candidate neighbor set after removing neighbors missing in the target view.	$\mathcal{N}_{i,v}$	Final reliable neighbor set of sample i in target view v.
$R_{i,j}^{(v)}$	Neighborhood reliability score of candidate neighbor j for reconstructing sample i in target view v.	$Z^{(v)}, d_e$	Embedding feature matrix of the completed v-th view ($\mathbb{R}^{n\times d_e}$) and the unified embedding dimension d_e.
Z^f	Fused feature representation of all views ($\mathbb{R}^{n\times d_e}$).	$T^{(v)}$	Transition probability matrix of the v-th view ($\mathbb{R}^{n\times n}$).
$\Psi^{(v)}$	High-order semantic structure matrix of the v-th view ($\mathbb{R}^{n\times n}$).	L	Label similarity matrix in the label space ($\mathbb{R}^{n\times n}$).
$S^{(v)}$	Feature similarity matrix induced by the v-th view ($[0,1]^{n\times n}$).	B	Predicted view-quality distribution ($[0,1]^{n\times m}$).
E	Trustworthy view-quality distribution ($[0,1]^{n\times m}$).	P	Final multi-label prediction matrix ($[0,1]^{n\times\ell}$).
$\mathbb{I}(\cdot), \|\cdot\|_2$	Indicator function and ℓ_2 norm.	$\odot$	Hadamard product.

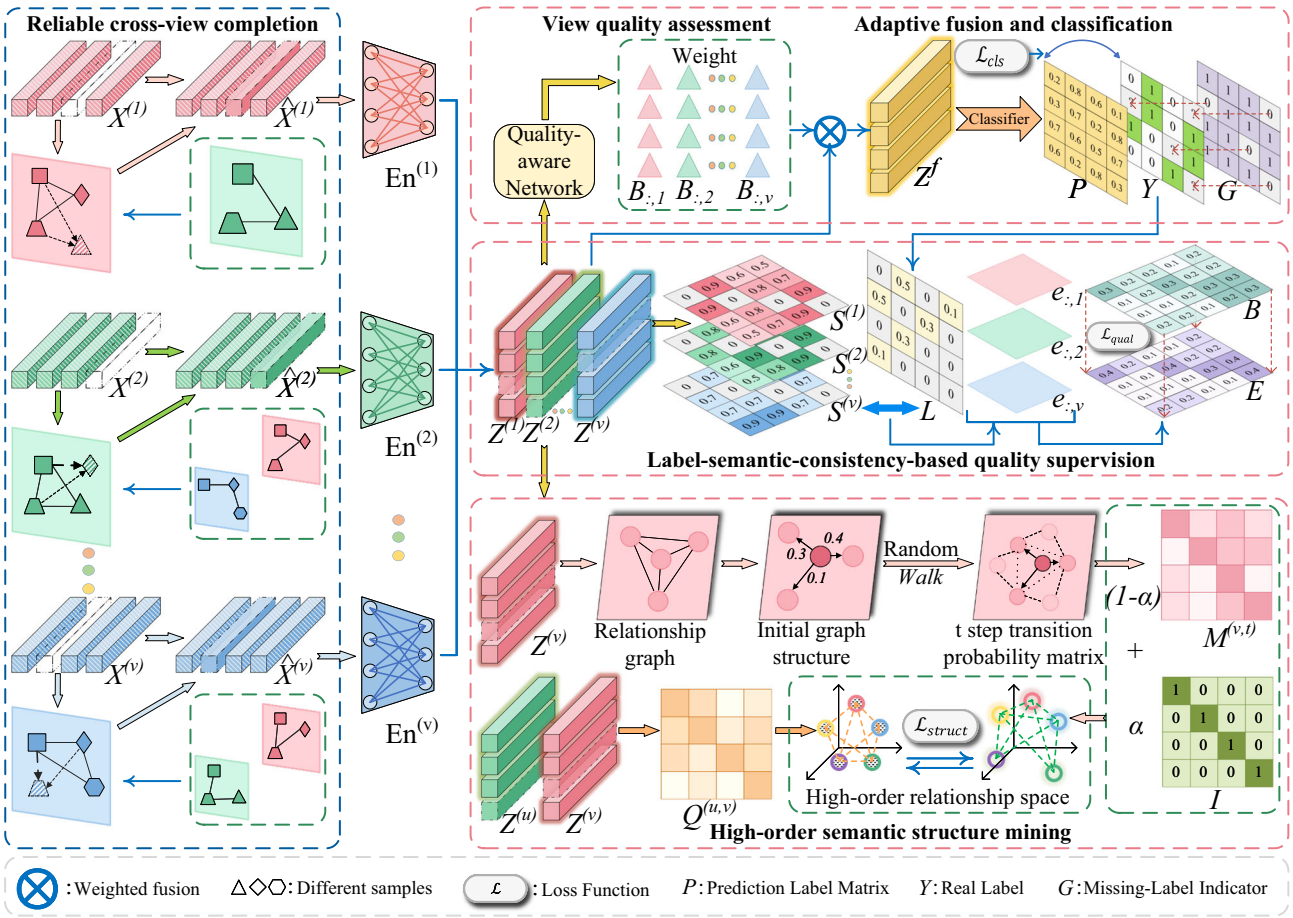


Fig. 1 Overall framework of ReCHSM. The incomplete views are first reconstructed and mapped into view-specific embeddings. High-order semantic structure mining and label-semantic-consistency-based view quality assessment are then performed in parallel to improve representation learning and estimate sample-wise view weights, respectively. The learned weights are finally used for adaptive multi-view fusion and multi-label classification. Different colors and shapes denote different views and samples, respectively

$\mathcal{L}_{\text{qual}}$. Finally, the sample-specific weights B adaptively fuse the view-specific embeddings into Z^f. The classifier then maps Z^f to the multi-label prediction matrix P. Thus, view completion provides reliable inputs, high-order structure mining improves the learned representations, and view quality assessment controls their contributions during final fusion.

3.3 View completion via reliable cross-view neighborhood relation transfer

The intrinsic challenge of incomplete multi-view data is that missing views reduce the completeness of sample representations and destroy the local neighborhood topology required for downstream representation learning. Directly discarding incomplete samples wastes the information retained in their observable views and may introduce sample-selection bias, whereas zero-padding creates artificial feature patterns that distort inter-sample distances. Therefore, appropriately reconstructing missing views before representation learning is important for preserving both feature information and local geometric structures. The cross-view relation transfer strategy introduced in CRTC (Wang et al. 2023) provides a topology-aware solution by transferring neighborhood relations preserved in observable source views to a missing target view. However, due to cross-view heterogeneity, different source views may provide inconsistent neighborhood evidence, and directly transferring their nearest neighbors may introduce unreliable candidates. We therefore adapt this strategy to the IMvPMLC setting by screening transferred candidates before reconstruction and using nonlinear refinement to suppress residual reconstruction noise. The overall workflow is illustrated in Fig. 2, and its detailed formulation is presented below.

First, we use Euclidean distance to evaluate the similarity between samples for identifying neighbors. For each view, we calculate a Euclidean distance matrix $D^{(v)} \in \mathbb{R}^{n\times n}$ over the observed sample set $\mathcal{I}_v = \{i \mid W_{i,v} = 1\}$. The elements

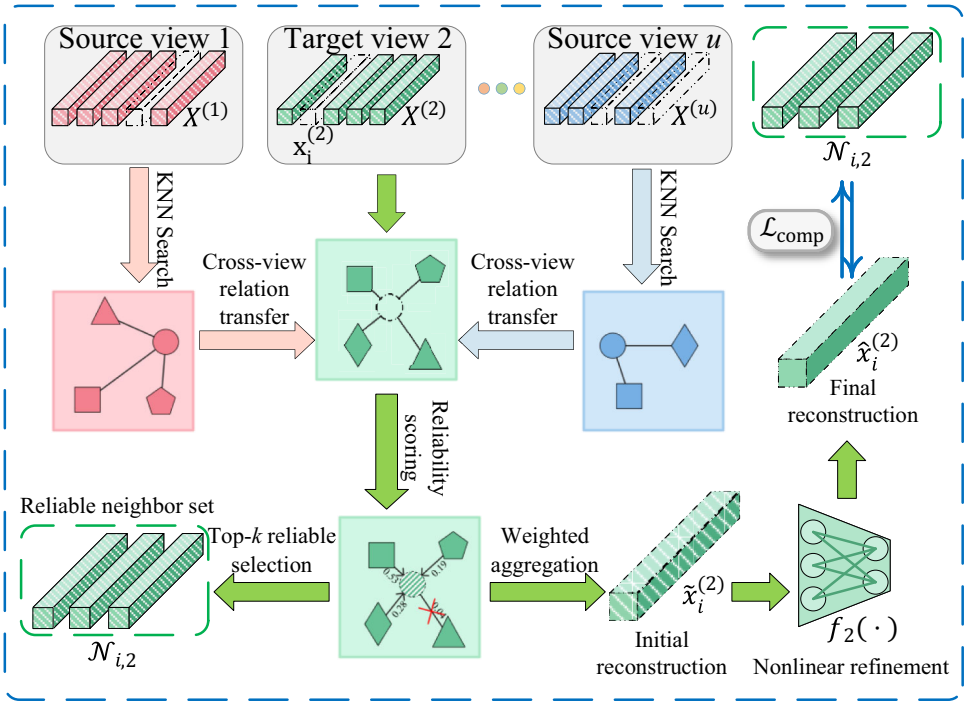


Fig. 2 Illustration of reliable cross-view completion for a missing sample i in view 2. Neighborhood relations identified in the observable source views are transferred to the target view, and unreliable candidates are removed through reliability scoring and top-k selection. The retained neighbors are then aggregated to obtain an initial reconstruction, which is further refined by a nonlinear network. The completion module is pre-trained using $\mathcal{L}_{\text{comp}}$

of this matrix are defined as:

$$D_{i,j}^{(v)} = \begin{cases} \|x_i^{(v)} - x_j^{(v)}\|_2, & \text{if } i, j \in \mathcal{I}_v, \\ +\infty, & \text{otherwise.} \end{cases} \tag{1}$$

Next, we transfer neighborhood relations from the source views to the target view to guide the completion process. For a sample i missing in target view v, we retrieve its top-k nearest neighbors from each observable source view $u \in \mathcal{V}_{\text{obs}}^{(i)} = \{u \mid W_{i,u} = 1\}$ according to the corresponding distance matrix $D^{(u)}$, and form a candidate neighbor set $\mathcal{P}_{i,u} = \{j_1, \ldots, j_k\}$. Some candidates in $\mathcal{P}_{i,u}$ may also be missing in the target view v. Such candidates cannot be directly used for reconstruction. We therefore remove them and obtain a transferred candidate neighbor set $\widetilde{\mathcal{P}}_{i,u}^{(v)}$, where $\left|\widetilde{\mathcal{P}}_{i,u}^{(v)}\right| \le k$. To measure the reliability of candidate j, we define its occurrence frequency $c_{i,j}^{(v)}$ and accumulated distance $\Delta_{i,j}^{(v)}$ across the observable source views as follows:

$$\begin{aligned} c_{i,j}^{(v)} &= \sum_{u \in \mathcal{V}_{\text{obs}}^{(i)}} \mathbb{I}\left(j \in \widetilde{\mathcal{P}}_{i,u}^{(v)}\right), \\ \Delta_{i,j}^{(v)} &= \sum_{u \in \mathcal{V}_{\text{obs}}^{(i)}} \mathbb{I}\left(j \in \widetilde{\mathcal{P}}_{i,u}^{(v)}\right) D_{i,j}^{(u)}. \end{aligned} \tag{2}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

To account for cross-view heterogeneity, we jointly evaluate candidate reliability using cross-view occurrence consistency and geometric proximity. The average source-view distance of candidate j is $\Delta_{i,j}^{(v)}/c_{i,j}^{(v)}$. We therefore define its reliability as the occurrence frequency divided by this average distance:

$$R_{i,j}^{(v)} = \frac{c_{i,j}^{(v)}}{\Delta_{i,j}^{(v)}/c_{i,j}^{(v)}} = \frac{\left(c_{i,j}^{(v)}\right)^2}{\Delta_{i,j}^{(v)}}. \tag{3}$$

Based on the neighborhood reliability score, we identify the top-k candidates from the union of all transferred candidate neighbor sets $\widetilde{\mathcal{P}}_{i,u}^{(v)}$ to form the target neighbor set $\mathcal{N}_{i,v}$:

$$\mathcal{N}_{i,v} = \text{TopK}_{j \in \bigcup_{u \in \mathcal{V}_{\text{obs}}^{(i)}} \widetilde{\mathcal{P}}_{i,u}^{(v)}} \left(R_{i,j}^{(v)}\right). \tag{4}$$

If fewer than k valid candidates are available, all available candidates are retained. If no valid candidate exists, no artificial neighbor is introduced, and the zero-initialized feature is passed to the subsequent nonlinear refinement network.

We then obtain an initial reconstruction of the missing feature $x_i^{(v)}$ by aggregating these neighbors with reliability-based weights:

$$\begin{aligned} \omega_{i,j} &= \frac{\exp\left(R_{i,j}^{(v)}\right)}{\sum_{k \in \mathcal{N}_{i,v}} \exp\left(R_{i,k}^{(v)}\right)}, \\ \tilde{x}_i^{(v)} &= \sum_{j \in \mathcal{N}_{i,v}} \omega_{i,j} x_j^{(v)}. \end{aligned} \tag{5}$$

where $\omega_{i,j}$ denotes the contribution weight of neighbor j.

Linear weighted reconstruction may not fully capture nonlinear semantics in high-dimensional feature spaces. We

therefore employ a lightweight multilayer-perceptron-based refinement network $f_v(\cdot)$ to refine the initially completed features. The final completed feature $\hat{x}_i^{(v)}$ is defined as

$$\hat{x}_i^{(v)} = \begin{cases} x_i^{(v)}, & \text{if } W_{i,v} = 1, \\ f_v(\tilde{x}_i^{(v)}), & \text{if } W_{i,v} = 0. \end{cases} \tag{6}$$

To encourage the completed features to remain close to the local subspaces formed by their reliable neighbors, we define a completion loss based on mean squared error. Specifically, the loss measures the discrepancies between each completed feature and its selected reliable neighbors:

$$\mathcal{L}_{\text{comp}} = \frac{1}{N_{\text{miss}}} \sum_{v=1}^{m} \sum_{i:W_{i,v}=0} \frac{1}{\max\{1, |\mathcal{N}_{i,v}|\}} \sum_{j \in \mathcal{N}_{i,v}} \left\| \hat{x}_i^{(v)} - x_j^{(v)} \right\|_2^2 \tag{7}$$

where $N_{\text{miss}} = \sum_{v=1}^{m} |\{i \mid W_{i,v} = 0\}|$ is the total number of missing samples across all views. When $\mathcal{N}_{i,v} = \varnothing$, the corresponding inner loss term is defined as zero.

After cross-view completion, we obtain a completed feature matrix $\hat{X}^{(v)} \in \mathbb{R}^{n \times d_v}$ for each view v. To transform heterogeneous view features into a shared latent representation space, we further employ a view-specific encoder $\text{En}^{(v)}(\cdot)$ to generate the embedding feature matrix:

$$Z^{(v)} = \text{En}^{(v)}\left(\hat{X}^{(v)}\right) \tag{8}$$

where $Z^{(v)} \in \mathbb{R}^{n \times d_e}$, and d_e denotes the unified embedding dimension. In implementation, each $\text{En}^{(v)}(\cdot)$ is a view-specific multilayer perceptron composed of stacked fully connected blocks with batch normalization and ReLU activation. Dropout is applied to the intermediate layers.

3.4 Random-walk-based high-order semantic structure mining

The completion module in Section 3.3 reconstructs the missing views and repairs the local neighborhood topology. However, these recovered relations mainly describe first-order proximity between directly connected samples. Such pairwise relations may be insufficient to characterize complex latent structures, since samples that are not immediate neighbors may still be semantically related through shared neighborhoods or multi-hop connection paths. Therefore, exploiting high-order neighborhood information is important for revealing deeper inter-sample dependencies and guiding the learning of more discriminative representations. Motivated by this observation, we introduce a random-walk-based high-order semantic structure mining module. Taking the completed view-specific embeddings as input, this module propagates the repaired local relations through multi-step transitions and transforms first-order neighborhoods into broader high-order semantic dependencies. The resulting high-order structures are then used as structural supervision for representation learning, as detailed below.

Based on the embedding feature $Z^{(v)}$ for the v-th view, we first construct the transition probability matrix $T^{(v)}$ using a Gaussian kernel:

$$T_{i,j}^{(v)} = \frac{\exp\left(-\frac{\|\tilde{z}_i^{(v)} - \tilde{z}_j^{(v)}\|_2^2}{2\sigma^2}\right)}{\sum_{k=1}^{n} \exp\left(-\frac{\|\tilde{z}_i^{(v)} - \tilde{z}_k^{(v)}\|_2^2}{2\sigma^2}\right)} \tag{9}$$

where $\tilde{z}_i^{(v)} = z_i^{(v)} / \|z_i^{(v)}\|_2$, and $z_i^{(v)}$ is the embedding of sample i in $Z^{(v)}$. The kernel bandwidth σ is set to 0.1. The i-th row of $T^{(v)} \in \mathbb{R}^{n \times n}$ represents the one-step transition probability distribution from sample i to all samples.

Then, to further capture multi-hop neighborhood relations, we perform a t-step random walk propagation:

$$\begin{cases} M^{(v,1)} = T^{(v)}, \\ M^{(v,t)} = M^{(v,t-1)} T^{(v)} = \left(T^{(v)}\right)^t, \quad t \geq 2. \end{cases} \tag{10}$$

where $M_{i,j}^{(v,t)}$ denotes the t-step transition strength from sample i to sample j. The step number t controls the propagation depth: increasing t captures broader multi-hop relations, whereas an excessively large value may cause over-diffusion.

To balance local and global information, we fuse $M^{(v,t)}$ with the identity matrix $I \in \mathbb{R}^{n \times n}$, yielding the final high-order semantic structure matrix $\Psi^{(v)}$:

$$\Psi^{(v)} = \alpha I + (1 - \alpha) M^{(v,t)} \tag{11}$$

where α is the balance coefficient between sample-specific identity information and the propagated high-order relations. A moderate α helps retain sample discrimination while exploiting multi-hop structural information. Based on the sensitivity results in Section 4.3, we set $t = 15$ and $\alpha = 0.5$.

Finally, we use the mined high-order associations as structural supervision for cross-view representation learning. For the u-th and v-th views, the cross-view similarity distribution $Q^{(u,v)} \in \mathbb{R}^{n \times n}$ is computed as follows:

$$Q_{i,j}^{(u,v)} = \frac{\exp\left(\langle \tilde{z}_i^{(u)}, \tilde{z}_j^{(v)} \rangle / \tau\right)}{\sum_{k=1}^{n} \exp\left(\langle \tilde{z}_i^{(u)}, \tilde{z}_k^{(v)} \rangle / \tau\right)} \tag{12}$$

where $\langle \tilde{z}_i^{(u)}, \tilde{z}_j^{(v)} \rangle$ denotes the inner product between two ℓ_2-normalized embeddings, which corresponds to cosine similarity, and τ is a temperature hyperparameter set to 0.2.

The structural constraint loss is then defined as

$$\mathcal{L}_{\text{struct}} = -\frac{1}{mn^2}\sum_{u,v=1}^{m}\sum_{i,j=1}^{n}\Big[\Psi_{i,j}^{(v)}\log Q_{i,j}^{(u,v)} + \big(1-\Psi_{i,j}^{(v)}\big)\log\big(1-Q_{i,j}^{(u,v)}\big)\Big]. \tag{13}$$

3.5 Label-semantic-consistency-based view quality assessment

Although the preceding modules reconstruct the missing views and enhance the structural semantics of the learned representations, the reliability of different views may still vary across samples because of cross-view heterogeneity and residual completion errors. Treating all views equally, or assigning a fixed weight to each view, cannot adequately capture such sample-specific quality variations and may allow less reliable representations to interfere with the final fusion (Liu et al. 2025). Therefore, it is necessary to evaluate view quality at the sample level and adaptively regulate the contribution of each view. To make this evaluation directly relevant to the downstream multi-label task, we assess view reliability according to label-semantic consistency. Specifically, a view is considered more reliable for a sample if the inter-sample relations induced by its representation are more consistent with the semantic structure characterized by the observed labels. Based on this principle, we construct a trustworthy quality distribution as the supervision signal and learn sample-wise adaptive view weights for subsequent fusion.

Specifically, we stack the embedding features $\{Z^{(v)}\}_{v=1}^{m}$ along the view dimension to form $C \in \mathbb{R}^{n\times m\times d_e}$, where $C_i = [z_i^{(1)}, z_i^{(2)}, \ldots, z_i^{(m)}]^\top \in \mathbb{R}^{m\times d_e}$ contains all view-specific embeddings of sample i. A quality-aware network g_ϕ, implemented as a three-layer perceptron with linear layers, ReLU activation, and Dropout, maps C to the predicted view-quality distribution $B \in [0,1]^{n\times m}$:

$$B = \text{softmax}\big(g_\phi(C)\big) \tag{14}$$

where the softmax operation is applied along the view dimension.

Then, to provide guidance for training the quality-aware network g_ϕ, we further construct a supervisory signal by measuring the consistency between label-induced semantic structure and view-induced similarity. To achieve this, we first define the label similarity matrix as follows:

$$L = \frac{(Y\odot G)(Y\odot G)^\top}{GG^\top} \tag{15}$$

where $\odot$ denotes the Hadamard product, and the division is performed element-wise. The matrix $GG^\top$ records the number of commonly observed labels for each sample pair and normalizes L, which represents pairwise semantic similarity in the observed label space.

Next, we use cosine similarity to characterize the relationship between embedded features:

$$S_{i,j}^{(v)} = \frac{1+\cos(z_i^{(v)}, z_j^{(v)})}{2} \tag{16}$$

where $z_i^{(v)}$ is the embedding of sample i in the v-th view, and $S^{(v)} \in [0,1]^{n\times n}$ is the corresponding feature similarity matrix. Here, $\cos(x,y) = x^\top y/(\|x\|_2\,\|y\|_2)$. The transformation $(1+\cos(\cdot,\cdot))/2$ maps cosine similarity from $[-1,1]$ to $[0,1]$, making it compatible with L.

To quantify how well each view preserves the semantic structure revealed by the observed labels, we measure the discrepancy between similarities derived from labels and those derived from features. For sample i in view v, we define the semantic consistency score as:

$$e_{i,v} = \frac{1}{n-1}\sum_{j=1,j\neq i}^{n}\Big(L_{i,j} - S_{i,j}^{(v)}\Big)^2 \tag{17}$$

A smaller $e_{i,v}$ indicates stronger agreement between the similarity structure induced by view v and the observed label semantics of sample i. Such a view is therefore considered more reliable for that sample. We then normalize the consistency scores across views to construct the trustworthy view-quality distribution $E \in [0,1]^{n\times m}$:

$$E_{i,v} = \frac{\exp(-e_{i,v})}{\sum_{k=1}^{m}\exp(-e_{i,k})} \tag{18}$$

In this way, the normalization assigns larger values in E to views with higher semantic consistency, enabling E to serve as a trustworthy target distribution for supervising quality prediction.

Finally, we align the predicted quality distribution $B \in [0,1]^{n\times m}$ with the reliable distribution $E \in [0,1]^{n\times m}$ using the following quality assessment loss:

$$\mathcal{L}_{\text{qual}} = \frac{1}{n}\sum_{i=1}^{n}\sum_{v=1}^{m}E_{i,v}\log\frac{E_{i,v}}{B_{i,v}} \tag{19}$$

Additionally, we employ a gradient isolation mechanism in this module. Specifically, we block gradients propagated through downstream fusion and classification from updating the quality-aware network and optimize this network only through $\mathcal{L}_{\text{qual}}$. This design allows the network to

focus on sample-wise view reliability under label-semantic-consistency supervision without interference from the classification objective.

3.6 Quality-aware adaptive fusion and classification

After the high-order structure branch improves the view-specific representations and the quality assessment branch estimates their sample-wise reliability, the predicted quality distribution B adaptively aggregates the embeddings $\{Z^{(v)}\}_{v=1}^{m}$. Views with higher estimated quality contribute more to the fused representation, whereas less reliable views contribute less:

$$Z^{f} = \sum_{v=1}^{m} B_{:,v} \odot Z^{(v)} \tag{20}$$

where $B_{:,v}$ denotes the learned sample-level quality weight vector for the v-th view, and $\odot$ denotes broadcasted element-wise multiplication along the feature dimension.

Subsequently, the classifier $h(\cdot)$ maps the unified feature representation Z^{f} to the final multi-label prediction matrix $P \in [0,1]^{n\times\ell}$. In implementation, $h(\cdot)$ consists of a LeakyReLU activation, a two-layer MLP, and a final Sigmoid activation function. To measure the discrepancy between the predictions and ground truth, we employ the binary cross-entropy loss:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{n\ell}\sum_{i=1}^{n}\sum_{j=1}^{\ell} G_{i,j}\Big[Y_{i,j}\log(P_{i,j}) + (1-Y_{i,j})\log(1-P_{i,j})\Big] \tag{21}$$

3.7 Total loss and model training

We adopt a two-stage strategy to optimize ReCHSM. During pre-training, we optimize the completion networks $\{f_v\}_{v=1}^{m}$ for $T_{\text{pre}} = 50$ epochs by minimizing $\mathcal{L}_{\text{comp}}$ in (7). We then freeze their parameters to provide stable completed features during the main training stage. We pre-train all completion networks exclusively on the training subset.

In the main training stage, the total loss $\mathcal{L}_{\text{total}}$ is formulated as a weighted combination of the classification loss $\mathcal{L}_{\text{cls}}$ in (21), the high-order semantic structure loss $\mathcal{L}_{\text{struct}}$ in (13), and the view quality assessment loss $\mathcal{L}_{\text{qual}}$ in (19):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \beta\mathcal{L}_{\text{struct}} + \gamma\mathcal{L}_{\text{qual}}, \tag{22}$$

where β and γ serve as penalty coefficients. During the first $T_{\text{warm}} = 20$ epochs, we set $\Psi^{(v)}$ to the identity matrix, so $\mathcal{L}_{\text{struct}}$ mainly encourages embeddings of the same sample from different views to remain close. This warm-up stabilizes the representation space before we introduce the high-order structures. Afterward, we replace $\Psi^{(v)}$ with the random-walk-based high-order semantic structure matrix. Algorithm 1 outlines the comprehensive training procedure of ReCHSM.

Algorithm 1 Training procedure of ReCHSM.

Input: Multi-view data $\{X^{(v)}\}_{v=1}^{m}$, missing-label indicator matrix G, missing-view indicator matrix W, and ground-truth label matrix Y.

Initialization: Randomly initialize refinement networks $\{f_v\}_{v=1}^{m}$, view-specific encoders $\{\text{En}^{(v)}\}_{v=1}^{m}$, quality-aware network g_ϕ, and classifier h; set hyperparameters α, β, γ, k, t, σ, τ, pre-training epochs T_{pre}, warm-up epochs T_{warm}, and main training epochs T_{main}.

Stage 1: Pre-training of completion module

for $r = 1$ **to** T_{pre} **do**

1. For each missing sample in target view v, identify reliable transferred neighbors $\mathcal{N}_{i,v}$ according to (2)–(4).
2. Reconstruct the initial missing features $\tilde{x}_i^{(v)}$ by weighted aggregation using (5), and refine them by $f_v(\cdot)$ according to (6).
3. Update the learnable parameters of $\{f_v\}_{v=1}^{m}$ by minimizing $\mathcal{L}_{\text{comp}}$ in (7).

end for

Freeze the learnable parameters of the completion module.

Stage 2: Main training with frozen completion module

for $r = 1$ **to** T_{main} **do**

1. Recover missing views using the frozen completion module to obtain $\hat{X}^{(v)}$, and generate view-specific embeddings $\{Z^{(v)}\}_{v=1}^{m}$ according to (8).
2. **if** $r \le T_{\text{warm}}$ **then** set $\Psi^{(v)} = I$ for all views; **else** construct $\Psi^{(v)}$ via random walk using (9)–(11).
3. Predict the sample-level view quality distribution B using (14), and derive the trustworthy distribution E using (15)–(18).
4. Fuse the multi-view embeddings into Z^{f} using (20), obtain predictions P, and compute $\mathcal{L}_{\text{qual}}$, $\mathcal{L}_{\text{struct}}$, and $\mathcal{L}_{\text{cls}}$.
5. Update $\{\text{En}^{(v)}\}_{v=1}^{m}$, g_ϕ, and h by minimizing $\mathcal{L}_{\text{total}}$.

end for

Output: Trained ReCHSM model.

4 Experiment

4.1 Experimental setup

4.1.1 Datasets

Following the experimental settings widely adopted in prior studies in this field (Jiang et al. 2026b; Li and Chen 2022; Liu et al. 2023c; Long et al. 2024), we evaluate ReCHSM on five benchmark datasets widely used in MvMLC, namely Corel5k, ESPGame, Pascal07, Mirflickr, and IAPRTC12. They cover diverse semantic scenarios, including natural scenes, object semantics, and social media images. Each dataset provides six heterogeneous view fea-

tures HSV, GIST, Hue, LAB, RGB, and SIFT, which describe samples from different perspectives such as color, texture, and spatial layout. Table 2 reports brief statistical information for these datasets. More detailed information is as follows:

(1) **Corel5k** (Duygulu et al. 2002): This dataset contains 4,999 images across 260 categories. Each image typically has 1 to 5 labels. This indicates significant label sparsity.
(2) **Pascal07** (Everingham et al. 2010): This dataset contains 9,963 images with 20 object categories. It is usually used for object detection and recognition.
(3) **ESPGame** (von Ahn and Dabbish 2004): Collected from an online collaborative game, this dataset includes 20,770 images and 268 labels. Each image has 1–15 descriptive labels.
(4) **IAPRTC12** (Escalante et al. 2010): This large-scale dataset comprises 19,627 high-quality natural scene images across 291 categories. Labels are extracted from captions or descriptions. Each sample can have up to 23 labels, showing high semantic complexity.
(5) **Mirflickr** (Huiskes and Lew 2008): This dataset contains 25,000 images from Flickr. It features people, animals, and landscapes. We use 38 core semantic labels. These labels represent the typical distribution of social photography.

4.1.2 Construction of dual incompleteness

To validate ReCHSM, we follow the protocols of previous studies (Jiang et al. 2026b; Li and Chen 2022; Liu et al. 2023c; Long et al. 2024) to construct datasets with incomplete views and labels. Specifically, we randomly remove 50% of the observations from each view while retaining at least one observed view for every sample. We simulate label incompleteness by randomly withholding 50% of the labels across all categories. We then divide the resulting data into training (70%), validation (15%), and test (15%) subsets. Although the feature matrices remain incomplete in all subsets, we retain complete ground-truth labels for the validation and test sets to support rigorous evaluation.

Table 2 Statistics of the benchmark datasets. The Avg. Labels denote the average number of semantic labels per sample

Dataset	Samples	Views	Labels	Avg. Labels
Corel5k	4,999	6	260	3.40
Pascal07	9,963	6	20	1.47
ESPGame	20,770	6	268	4.69
IAPRTC12	19,627	6	291	5.72
Mirflickr	25,000	6	38	4.72

4.1.3 Compared methods

To ascertain the capability of ReCHSM on the IMvPMLC task, we compare it with nine established models: DSPNF (Jiang et al. 2026b), NAIM3L (Li and Chen 2022), DICNet (Liu et al. 2023c), RANK (Liu et al. 2025), MSLPP (Long et al. 2024), iMvWL (Tan et al. 2018), C2AE (Yeh et al. 2017), CDMM (Zhao et al. 2021b), and LVSL (Zhao et al. 2023). Among these baselines, DSPNF, NAIM3L, DICNet, RANK, MSLPP, iMvWL, LVSL, and CDMM are representative MvMLC or IMvPMLC methods closely related to the task studied in this paper. C2AE is a classic deep autoencoder-based model for multi-label classification that improves prediction performance by jointly learning feature representations and label embeddings, primarily for single-view scenarios. Table 3 summarizes the characteristics of the compared methods. Some baselines do not natively support the IMvPMLC task. For the single-view baseline C2AE, we evaluate each view separately and use the best result across all views for comparison. For C2AE, CDMM, and LVSL, which cannot directly process missing views, we replace unavailable view features with the average representation computed from the observed views. For LVSL and CDMM, which cannot directly handle missing labels, we impute unobserved label entries with zeros.

4.1.4 Evaluation metrics

In line with the methodologies in Jiang et al. (2026b); Li and Chen (2022); Liu et al. (2023c); Long et al. (2024), our performance assessment relies on four foundational metrics: Ranking Loss (RL), Average Precision (AP), Hamming Loss (HL), and the adapted Area Under the Curve (AUC). To maintain a monotonic evaluation scale where larger values consistently denote better performance, we employ the inverted forms $1 - \text{HL}$ and $1 - \text{RL}$ in place of the original loss values. Similarly, for comprehensive data evaluation, One-error (OE) and Coverage (Cov) are introduced in their complementary versions ($1 - \text{OE}$ and $1 - \text{Cov}$). This standardization allows for a more intuitive comparison, as any increase in the reported scores corresponds to an enhancement in model capability.

4.1.5 Implementation details

ReCHSM was implemented in Python using PyTorch. The main training stage employed SGD with a batch size of 128 and a momentum of 0.9. The learning rates were set to 0.005 for Pascal07, 0.5 for IAPRTC12, and 0.9 for the other datasets, while the completion module used a fixed learning rate of 0.001 on all datasets. The main performance experiments were conducted on a workstation with an Intel Core i5-12450H processor, 16 GB of RAM, and an NVIDIA

Table 3 Overview of the compared methods. "Multi-view" indicates whether a method is built for multi-view learning, whereas "Missing-view" and "Missing-label" indicate the capability to deal with incomplete data

Method	Source	Multi-view	Missing-view	Missing-label
C2AE	AAAI'17	✗	✗	✓
CDMM	KBS'20	✓	✗	✗
LVSL	TMM'22	✓	✗	✗
iMvWL	IJCAI'18	✓	✓	✓
NAIM3L	TPAMI'22	✓	✓	✓
DICNet	AAAI'23	✓	✓	✓
MSLPP	Neural Netw'24	✓	✓	✓
RANK	TPAMI'25	✓	✓	✓
DSPNF	ICONIP'25	✓	✓	✓

GeForce RTX 4050 Laptop GPU. The efficiency experiments in Section 4.5 were conducted on another workstation with an AMD Ryzen 9 7945HX processor, 32 GB of RAM, and an NVIDIA GeForce RTX 4060 Laptop GPU. All methods in the efficiency comparison were evaluated on the same RTX 4060 platform.

4.2 Experimental results and analysis

To comprehensively evaluate ReCHSM, we consider four main settings. First, we compare ReCHSM with nine competitive baselines under the dual-incomplete setting, which is the primary task studied in this paper, and report the results in Table 4. Furthermore, to further clarify how each type of incompleteness contributes to the performance, we evaluate two single-sided incomplete scenarios: complete views with 50% missing labels and 50% missing views with complete labels. The results are reported in Table 5. Third, we further explore the intrinsic stability of ReCHSM under varying missing rates, with the results presented in Fig. 3. Finally, we extend our evaluation to two complete datasets to further validate the generalization capability of ReCHSM, as illustrated in Fig. 4. Based on the experimental results presented above, the following key observations can be drawn:

- Table 4 shows that ReCHSM achieves the best or tied-best result on all four principal metrics across the five datasets. In particular, its AP reaches 0.583 on Pascal07 and 0.449 on Corel5k, exceeding the corresponding second-best results by 2.8 and 1.9 percentage points, respectively. These results demonstrate its effectiveness in learning discriminative multi-label representations under dual incompleteness.
- As shown in Table 5, ReCHSM achieves the best or tied-best result on almost all datasets and metrics in both single-incomplete settings. These results show that ReCHSM is effective under both dual and single incompleteness. Moreover, most methods perform better with missing labels than with missing views, indicating that view absence causes greater degradation by directly reducing the feature information available for representation learning. This finding supports the need for dedicated view recovery.
- As shown in Fig. 3, performance decreases as either missing rate increases, but the overall degradation remains moderate. ReCHSM is more sensitive to missing views than to missing labels, consistent with Table 5. This result further confirms the importance of preserving and recovering view information.
- As shown in Fig. 4, ReCHSM still achieves superior or highly competitive performance on most evaluation metrics. This observation indicates that the proposed framework is not limited to incomplete-data scenarios. Under complete-data conditions, the subsequent semantic modeling and adaptive fusion mechanisms remain effective, enabling the model to preserve strong discriminative ability on complete datasets.
- Through analysis and comparison across matched datasets, metrics, and baselines, ReCHSM shows its largest overall advantage under dual incompleteness. At the dataset level, the gains are most pronounced on Pascal07 under the dual-incomplete and missing-view settings and on Corel5k under the missing-label setting. This pattern suggests that ReCHSM benefits datasets with complementary heterogeneous views and sparse high-dimensional label spaces.
- As an additional robustness check, we evaluated ReCHSM under four non-random missing mechanisms, including view-specific missing rates, block-wise missing views, label-dependent missingness, and the one-view-observed setting. ReCHSM achieves the best or tied-best results on Corel5k and Pascal07. Detailed protocols, complete quantitative results, and analyses are provided in Supplementary Section S2 and Table S2.

Table 4 Experimental results of ten methods on five datasets under 50% missing views, 50% missing labels, and 70% training samples

DATA	METRIC	C2AE	iMvWL	NAIM3L	LVSL	CDMM	DICNet	MSLPP	RANK	DSPNF	OURS
Corel5k	AP↑	$0.227_{0.008}$	$0.283_{0.008}$	$0.309_{0.004}$	$0.342_{0.004}$	$0.354_{0.004}$	$0.381_{0.004}$	$0.413_{0.008}$	$0.427_{0.006}$	$\underline{0.430}_{0.007}$	$\mathbf{0.449}_{0.008}$
	1-HL↑	$0.980_{0.002}$	$0.978_{0.000}$	$\underline{0.987}_{0.000}$	$\underline{0.987}_{0.000}$	$\underline{0.987}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$
	1-RL↑	$0.804_{0.010}$	$0.865_{0.005}$	$0.878_{0.002}$	$0.881_{0.003}$	$0.884_{0.003}$	$0.882_{0.004}$	$0.901_{0.003}$	$\underline{0.913}0.002$	$0.904_{0.006}$	$\mathbf{0.916}_{0.003}$
	AUC↑	$0.806_{0.010}$	$0.868_{0.005}$	$0.881_{0.002}$	$0.884_{0.003}$	$0.888_{0.002}$	$0.884_{0.004}$	$0.903_{0.004}$	$\underline{0.916}_{0.003}$	$0.907_{0.006}$	$\mathbf{0.918}_{0.003}$
Pascal07	AP↑	$0.485_{0.008}$	$0.437_{0.018}$	$0.488_{0.003}$	$0.504_{0.005}$	$0.508_{0.005}$	$0.505_{0.012}$	$0.544_{0.010}$	$\underline{0.555}_{0.010}$	$0.553_{0.007}$	$\mathbf{0.583}_{0.010}$
	1-HL↑	$0.908_{0.002}$	$0.882_{0.004}$	$0.928_{0.001}$	$0.930_{0.000}$	$0.931_{0.001}$	$0.929_{0.001}$	$0.932_{0.001}$	$\underline{0.933}_{0.001}$	$\underline{0.933}_{0.001}$	$\mathbf{0.935}_{0.001}$
	1-RL↑	$0.745_{0.009}$	$0.736_{0.015}$	$0.783_{0.001}$	$0.806_{0.003}$	$0.812_{0.004}$	$0.783_{0.008}$	$0.819_{0.006}$	$\underline{0.830}_{0.008}$	$0.829_{0.004}$	$\mathbf{0.847}_{0.005}$
	AUC↑	$0.765_{0.010}$	$0.767_{0.015}$	$0.811_{0.001}$	$0.832_{0.002}$	$0.838_{0.003}$	$0.809_{0.006}$	$0.841_{0.004}$	$\underline{0.851}_{0.006}$	$0.849_{0.004}$	$\mathbf{0.868}_{0.004}$
ESPGame	AP↑	$0.202_{0.006}$	$0.244_{0.005}$	$0.246_{0.002}$	$0.285_{0.003}$	$0.289_{0.003}$	$0.297_{0.002}$	$0.310_{0.004}$	$0.314_{0.004}$	$\underline{0.321}_{0.005}$	$\mathbf{0.331}_{0.005}$
	1-HL↑	$0.971_{0.002}$	$\underline{0.972}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$
	1-RL↑	$0.772_{0.006}$	$0.808_{0.002}$	$0.818_{0.002}$	$0.829_{0.001}$	$0.832_{0.001}$	$0.832_{0.001}$	$0.843_{0.002}$	$0.848_{0.002}$	$\underline{0.851}_{0.002}$	$\mathbf{0.860}_{0.002}$
	AUC↑	$0.777_{0.006}$	$0.813_{0.002}$	$0.824_{0.002}$	$0.833_{0.002}$	$0.836_{0.001}$	$0.836_{0.001}$	$0.847_{0.002}$	$0.853_{0.002}$	$\underline{0.855}_{0.002}$	$\mathbf{0.863}_{0.002}$
IAPRTC12	AP↑	$0.224_{0.007}$	$0.237_{0.003}$	$0.261_{0.001}$	$0.304_{0.004}$	$0.305_{0.004}$	$0.323_{0.001}$	$0.340_{0.005}$	$0.346_{0.004}$	$\underline{0.355}_{0.005}$	$\mathbf{0.362}_{0.005}$
	1-HL↑	$0.965_{0.002}$	$0.969_{0.000}$	$\underline{0.980}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$
	1-RL↑	$0.806_{0.005}$	$0.833_{0.002}$	$0.848_{0.001}$	$0.861_{0.002}$	$0.862_{0.002}$	$0.873_{0.001}$	$0.882_{0.002}$	$0.888_{0.002}$	$\underline{0.892}_{0.003}$	$\mathbf{0.898}_{0.002}$
	AUC↑	$0.807_{0.005}$	$0.835_{0.001}$	$0.850_{0.001}$	$0.863_{0.001}$	$0.864_{0.002}$	$0.874_{0.000}$	$0.883_{0.002}$	$0.889_{0.002}$	$\underline{0.893}_{0.003}$	$\mathbf{0.899}_{0.001}$
Mirflickr	AP↑	$0.505_{0.008}$	$0.490_{0.012}$	$0.551_{0.002}$	$0.553_{0.002}$	$0.570_{0.002}$	$0.589_{0.005}$	$\underline{0.615}_{0.004}$	$0.602_{0.007}$	$0.611_{0.006}$	$\mathbf{0.626}_{0.003}$
	1-HL↑	$0.853_{0.004}$	$0.839_{0.002}$	$0.882_{0.001}$	$0.885_{0.001}$	$0.886_{0.001}$	$0.888_{0.002}$	$\underline{0.892}_{0.001}$	$0.890_{0.001}$	$\underline{0.892}_{0.001}$	$\mathbf{0.893}_{0.001}$
	1-RL↑	$0.821_{0.003}$	$0.803_{0.008}$	$0.844_{0.001}$	$0.856_{0.001}$	$0.856_{0.001}$	$0.863_{0.004}$	$\underline{0.879}_{0.002}$	$0.875_{0.003}$	$0.876_{0.003}$	$\mathbf{0.883}_{0.002}$
	AUC↑	$0.810_{0.004}$	$0.787_{0.012}$	$0.837_{0.001}$	$0.844_{0.001}$	$0.846_{0.001}$	$0.849_{0.004}$	$\underline{0.865}_{0.002}$	$0.860_{0.003}$	$0.863_{0.003}$	$\mathbf{0.869}_{0.002}$

Bold values indicate the best performance, while underlined values indicate the second-best performance in each comparison.

Table 5 Results under complete views with 50% missing labels and 50% missing views with complete labels

DATA	METRIC	Complete views and 50% missing labels				50% missing views and complete labels			
		iMvWL	NAIM3L	MSLPP	OURS	iMvWL	NAIM3L	MSLPP	OURS
Corel5k	AP↑	$0.308_{0.006}$	$0.319_{0.004}$	$\underline{0.443}_{0.009}$	$\mathbf{0.503}_{0.010}$	$0.289_{0.009}$	$0.317_{0.003}$	$\underline{0.443}_{0.011}$	$\mathbf{0.482}_{0.007}$
	1-HL↑	$0.979_{0.000}$	$0.987_{0.000}$	$\underline{0.988}_{0.000}$	$\mathbf{0.989}_{0.000}$	$0.979_{0.000}$	$\underline{0.987}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$
	1-RL↑	$0.880_{0.003}$	$0.886_{0.003}$	$\underline{0.912}_{0.004}$	$\mathbf{0.933}_{0.004}$	$0.870_{0.003}$	$0.885_{0.003}$	$\underline{0.912}_{0.003}$	$\mathbf{0.926}_{0.004}$
	AUC↑	$0.883_{0.003}$	$0.888_{0.003}$	$\underline{0.914}_{0.004}$	$\mathbf{0.935}_{0.003}$	$0.873_{0.003}$	$0.888_{0.002}$	$\underline{0.914}_{0.003}$	$\mathbf{0.929}_{0.004}$
Pascal07	AP↑	$0.470_{0.006}$	$0.493_{0.003}$	$\underline{0.611}_{0.009}$	$\mathbf{0.627}_{0.009}$	$0.435_{0.009}$	$0.493_{0.002}$	$\underline{0.565}_{0.011}$	$\mathbf{0.601}_{0.009}$
	1-HL↑	$0.888_{0.001}$	$0.929_{0.001}$	$\underline{0.937}_{0.001}$	$\mathbf{0.939}_{0.001}$	$0.882_{0.003}$	$0.929_{0.001}$	$\underline{0.934}_{0.001}$	$\mathbf{0.936}_{0.001}$
	1-RL↑	$0.763_{0.006}$	$0.790_{0.002}$	$\underline{0.864}_{0.005}$	$\mathbf{0.873}_{0.004}$	$0.737_{0.005}$	$0.791_{0.001}$	$\underline{0.835}_{0.007}$	$\mathbf{0.856}_{0.005}$
	AUC↑	$0.792_{0.006}$	$0.818_{0.001}$	$\underline{0.879}_{0.005}$	$\mathbf{0.890}_{0.004}$	$0.768_{0.006}$	$0.818_{0.001}$	$\underline{0.855}_{0.006}$	$\mathbf{0.875}_{0.005}$
ESPGame	AP↑	$0.258_{0.003}$	$0.249_{0.002}$	$\mathbf{0.366}_{0.002}$	$\underline{0.365}_{0.006}$	$0.245_{0.004}$	$0.249_{0.002}$	$\underline{0.329}_{0.005}$	$\mathbf{0.342}_{0.004}$
	1-HL↑	$\underline{0.972}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\underline{0.972}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$
	1-RL↑	$0.815_{0.002}$	$0.822_{0.001}$	$\underline{0.866}_{0.001}$	$\mathbf{0.877}_{0.003}$	$0.810_{0.002}$	$0.822_{0.002}$	$\underline{0.854}_{0.002}$	$\mathbf{0.863}_{0.002}$
	AUC↑	$0.820_{0.002}$	$0.827_{0.001}$	$\underline{0.870}_{0.002}$	$\mathbf{0.880}_{0.003}$	$0.816_{0.002}$	$0.828_{0.002}$	$\underline{0.857}_{0.002}$	$\mathbf{0.866}_{0.002}$
IAPRTC12	AP↑	$0.250_{0.002}$	$0.265_{0.001}$	$\underline{0.408}_{0.004}$	$\mathbf{0.429}_{0.005}$	$0.234_{0.005}$	$0.265_{0.001}$	$\underline{0.360}_{0.004}$	$\mathbf{0.380}_{0.003}$
	1-HL↑	$0.969_{0.000}$	$\underline{0.981}_{0.000}$	$\mathbf{0.982}_{0.000}$	$\mathbf{0.982}_{0.000}$	$\underline{0.968}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$
	1-RL↑	$0.841_{0.002}$	$0.851_{0.001}$	$\underline{0.907}_{0.002}$	$\mathbf{0.919}_{0.001}$	$0.832_{0.003}$	$0.851_{0.001}$	$\underline{0.891}_{0.002}$	$\mathbf{0.902}_{0.002}$
	AUC↑	$0.842_{0.002}$	$0.853_{0.001}$	$\underline{0.907}_{0.002}$	$\mathbf{0.920}_{0.001}$	$0.835_{0.002}$	$0.853_{0.001}$	$\underline{0.892}_{0.002}$	$\mathbf{0.902}_{0.002}$
Mirflickr	AP↑	$0.502_{0.020}$	$0.554_{0.001}$	$\underline{0.668}_{0.004}$	$\mathbf{0.691}_{0.004}$	$0.490_{0.010}$	$0.553_{0.002}$	$\underline{0.625}_{0.004}$	$\mathbf{0.627}_{0.004}$
	1-HL↑	$0.843_{0.006}$	$0.882_{0.001}$	$\underline{0.902}_{0.001}$	$\mathbf{0.907}_{0.001}$	$0.863_{0.002}$	$0.882_{0.001}$	$\underline{0.894}_{0.001}$	$\mathbf{0.895}_{0.001}$
	1-RL↑	$0.812_{0.020}$	$0.846_{0.001}$	$\underline{0.902}_{0.002}$	$\mathbf{0.911}_{0.002}$	$0.806_{0.007}$	$0.846_{0.001}$	$\underline{0.883}_{0.002}$	$\mathbf{0.885}_{0.002}$
	AUC↑	$0.795_{0.030}$	$0.839_{0.001}$	$\underline{0.885}_{0.002}$	$\mathbf{0.894}_{0.002}$	$0.797_{0.012}$	$0.838_{0.001}$	$\underline{0.869}_{0.002}$	$\mathbf{0.870}_{0.002}$

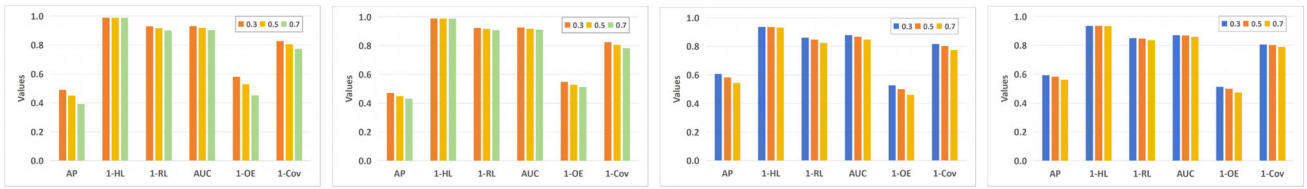
Bold values indicate the best performance, while underlined values indicate the second-best performance in each comparison.

4.3 Parameter study

To investigate the sensitivity of ReCHSM to its primary hyperparameters, we conduct a parameter study focusing on the penalty coefficients β and γ, the number of nearest neighbors k, the random-walk step number t, and the balance coefficient α. These analyses are performed on Corel5k and Pascal07 under 50% missing views, 50% missing labels, and a 70% training ratio.

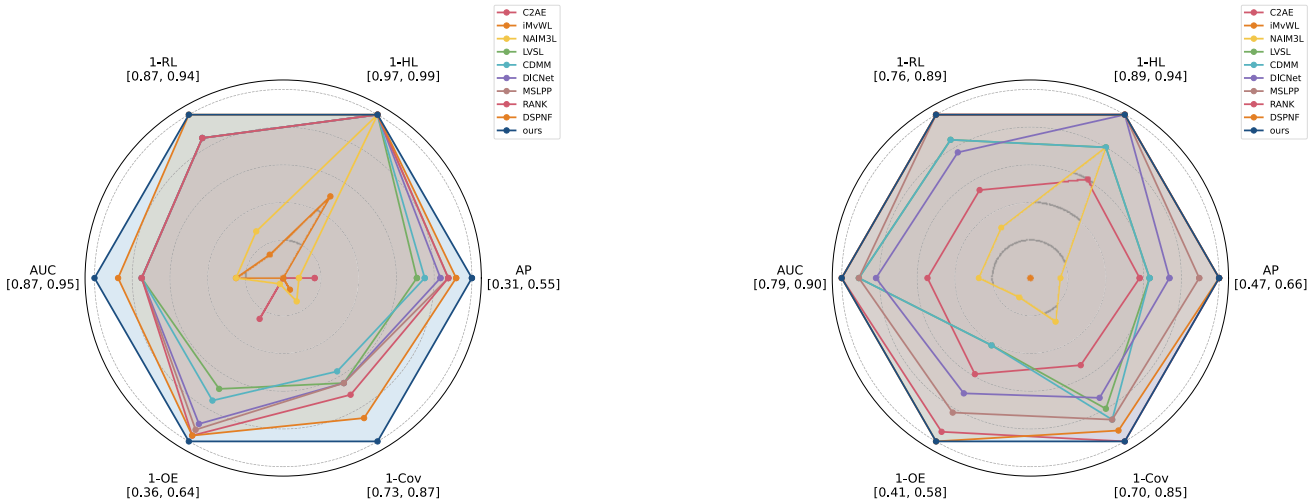
Regarding β and γ: Both parameters are selected from $\{1e{-}2, 1e{-}1, 5e{-}1, 1e0, 1e1\}$, and the results are shown in Figs. 5a and 5b, respectively. The results show that ReCHSM remains relatively stable for most tested combinations. However, setting γ to $1e1$ causes a marked performance decrease on Corel5k. This indicates that an excessively large γ may overemphasize the corresponding loss term and impair the balance of the overall objective. We therefore select values outside this extreme region.

Regarding k: The parameter k determines the initial candidate neighborhoods. As shown in Figs. 6a and 6b, ReCHSM remains stable for k from 5 to 15. The best AP is obtained at $k = 5$ on Corel5k and $k = 9$ on Pascal07, and these values are used in the subsequent experiments. The lim-



(a) Varying views (Corel5k) (b) Varying labels (Corel5k) (c) Varying views (Pascal07) (d) Varying labels (Pascal07)

Fig. 3 Robustness of ReCHSM under different missing rates. (**a**) and (**c**) illustrate the impact of varying missing views at a constant 50% missing-label rate. (**b**) and (**d**) show the impact of varying missing labels at a constant 50% missing-view rate. All metrics are presented in a higher-is-better form



(a) Corel5k (b) Pascal07

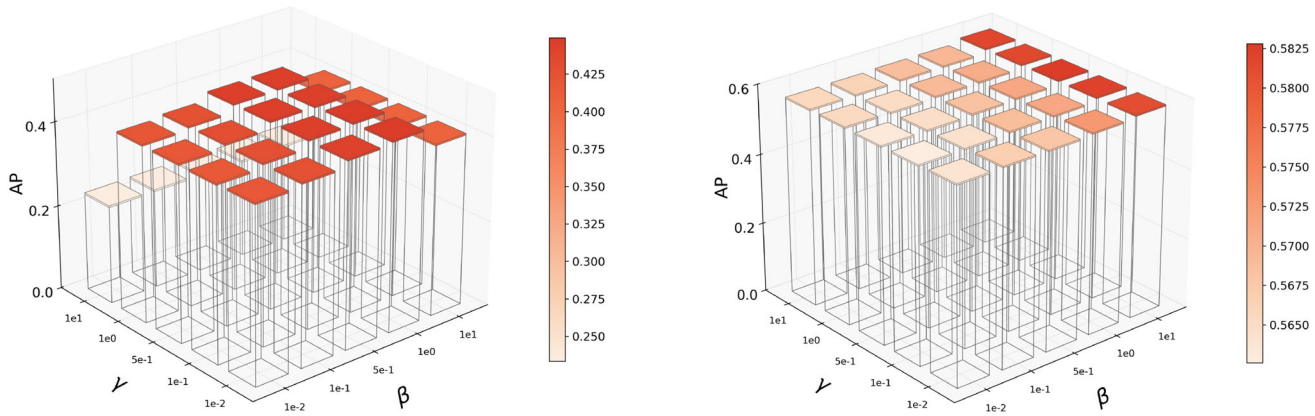
Fig. 4 Performance comparison of ten methods on two fully observed datasets. For each radar chart, performance improves as data points move from the center (worst result) toward the outer boundary (optimal result)

ited sensitivity can be attributed to reliability-based filtering and nonlinear refinement after candidate retrieval.

Regarding t and α: We vary t in {5, 10, 15, 20, 25} and vary α in {0.1, 0.3, 0.5, 0.7, 0.9}. The results are shown in Figs. 6c and 6d. For t, the highest AP is obtained at $t = 15$, suggesting that this value provides a suitable propagation depth. For α, the highest AP is obtained at $\alpha = 0.5$. This intermediate value provides a suitable balance between identity preservation and structural propagation. Moreover, the limited performance variations across the tested values indicate that ReCHSM is generally robust to moderate changes in both t and α.

4.4 Ablation study

We evaluate the contribution of each ReCHSM module under 50% missing views and labels. Full ReCHSM integrates four key modules and their corresponding loss functions: the classification backbone ($\mathcal{L}_{\text{cls}}$), the view completion module (Comp.) with $\mathcal{L}_{\text{comp}}$, the high-order semantic structure mining ($\mathcal{L}_{\text{struct}}$), and the label-semantic-consistency-based view quality assessment ($\mathcal{L}_{\text{qual}}$). The model trained only with $\mathcal{L}_{\text{cls}}$ is used as the baseline (Backbone). Table 6 reports the quantitative results on Corel5k and Pascal07. We also test severe view incompleteness by increasing the missing-view rate to 70%, while retaining 50% missing labels and a 70% training



(a) Corel5k (β and γ) (b) Pascal07 (β and γ)

Fig. 5 Hyperparameter study on β and γ for the Corel5k and Pascal07 datasets

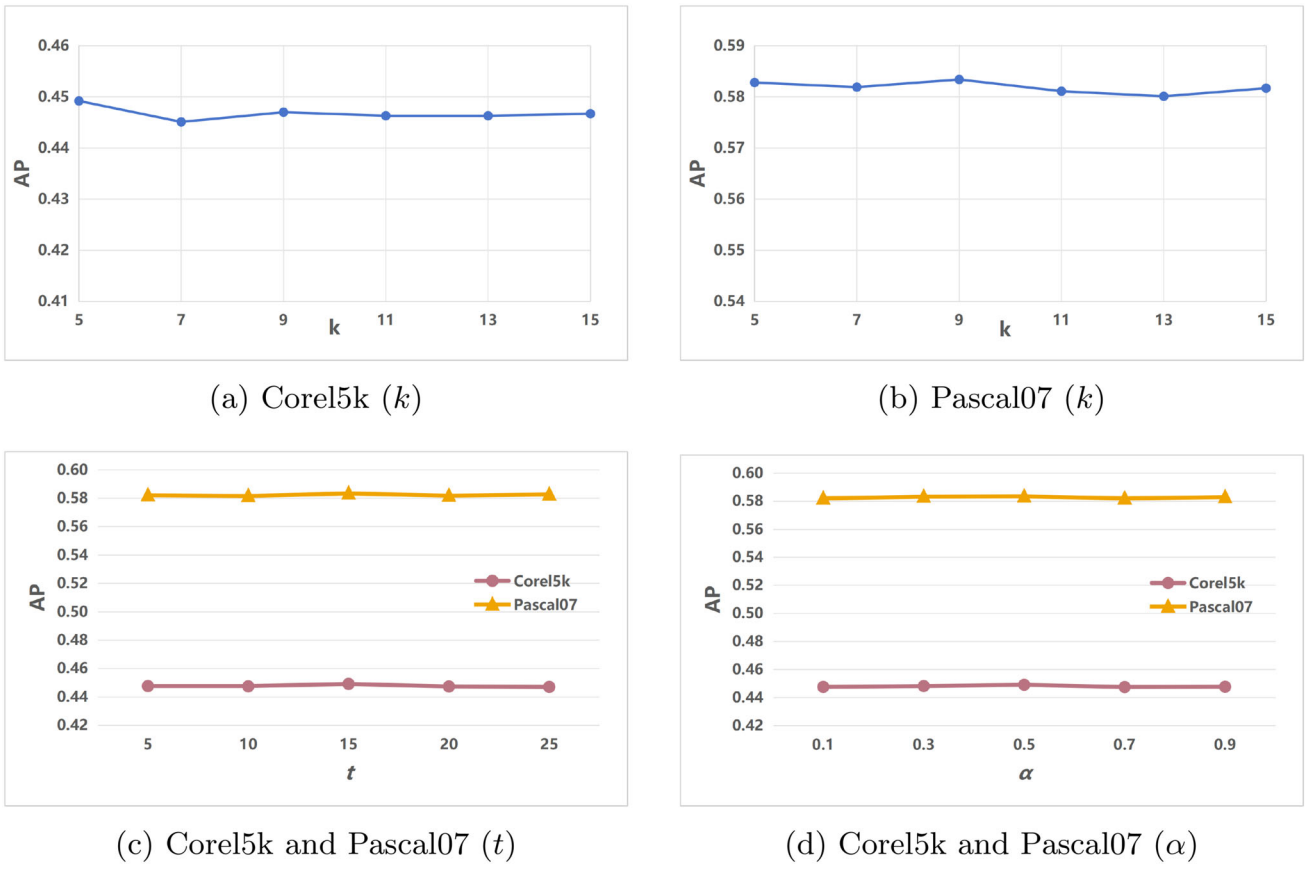


Fig. 6 Sensitivity of ReCHSM to k, t, and α on the Corel5k and Pascal07 datasets

ratio. Table 7 compares the full model with its variant without completion.

- **Effectiveness of reliable cross-view neighborhood relation transfer completion.** Compared with the Backbone, introducing Comp. improves AP from 0.392 to 0.420 on Corel5k and from 0.552 to 0.564 on Pascal07, corresponding to gains of 2.8 and 1.2 percentage points. This confirms that the completion strategy recovers useful missing-view information for downstream classification.
- **Synergistic effect of view completion and high-order semantic structure mining.** On Corel5k, combining Comp. with $\mathcal{L}_{\text{struct}}$ yields an AP of 0.447, which is 5.5 percentage points above the Backbone and higher than using Comp. (0.420) or $\mathcal{L}_{\text{struct}}$ (0.428) alone. Pascal07 shows the same trend. These results confirm that high-order structure mining extends repaired local relations to

Table 6 Performance results of the proposed components on Corel5k and Pascal07 under the dual-incomplete setting. *Comp.* represents the view completion module, and $\mathcal{L}_{\text{comp}}$ represents the loss function of this module

Method	Loss Functions				Corel5k				Pascal07			
	$\mathcal{L}_{\text{cls}}$	$\mathcal{L}_{\text{comp}}$	$\mathcal{L}_{\text{struct}}$	$\mathcal{L}_{\text{qual}}$	AP↑	1-HL↑	1-RL↑	AUC↑	AP↑	1-HL↑	1-RL↑	AUC↑
Backbone	✓				0.392	0.988	0.889	0.892	0.552	0.932	0.826	0.849
w/o Comp.	✓		✓		0.428	0.988	0.908	0.910	0.567	0.933	0.839	0.861
	✓			✓	0.392	0.987	0.892	0.895	0.553	0.932	0.827	0.850
	✓		✓	✓	0.428	0.988	0.908	0.910	0.566	0.934	0.839	0.860
Comp.	✓	✓			0.420	0.988	0.903	0.906	0.564	0.933	0.836	0.856
	✓	✓	✓		0.447	0.988	0.916	**0.919**	0.579	0.935	0.846	0.867
	✓	✓		✓	0.415	0.988	0.899	0.902	0.566	0.933	0.837	0.856
	✓	✓	✓	✓	**0.449**	**0.988**	**0.916**	0.918	**0.583**	**0.935**	**0.847**	**0.868**

Bold values indicate the best performance

Table 7 Ablation results of the view completion module on Corel5k and Pascal07 under 70% missing views, 50% missing labels, and 70% training samples

Method	Corel5k				Pascal07			
	AP↑	1-HL↑	1-RL↑	AUC↑	AP↑	1-HL↑	1-RL↑	AUC↑
w/o Comp.	0.365	0.987	0.885	0.888	0.525	0.930	0.809	0.835
Full model	0.392	0.987	0.900	0.902	0.543	0.931	0.824	0.847

broader semantic dependencies and complements view completion.

- **Effect of the view quality assessment module.** Used alone, this module performs close to the Backbone, indicating a limited standalone contribution. When added to completion and high-order structure mining, AP increases from 0.447 to 0.449 on Corel5k and from 0.579 to 0.583 on Pascal07, while the other metrics remain competitive. Thus, $\mathcal{L}_{\text{qual}}$ mainly provides auxiliary supervision after representation completeness and structural consistency have been improved.
- **Robustness of view completion under severe view incompleteness.** As shown in Table 7, the full model consistently outperforms the variant without view completion on both datasets. This indicates that the completion module continues to provide useful information even when most views are unavailable, confirming the robustness of the proposed completion strategy under severe view incompleteness.

To further examine the behavior of the completion mechanism, additional post-hoc diagnostics of candidate availability, accumulated-distance distribution, reconstruction-weight concentration, and effective neighbor number on Corel5k and Pascal07 are provided in Supplementary Section S1 and Table S1.

4.5 Complexity analysis

We analyze the computational requirements of ReCHSM in terms of time and space complexity. The major computational costs arise from reliable cross-view completion, view-specific representation learning, high-order semantic structure mining, and loss calculation. Let D_v denote the maximum number of neurons in the hidden layers associated with the v-th view, and let b denote the batch size.

For reliable cross-view completion, constructing the pairwise Euclidean distance matrices over all views requires $\mathcal{O}(mb^2 d_{\max})$, where m is the number of views and $d_{\max} = \max_v d_v$ is the maximum input dimension. The sorting-based KNN retrieval requires $\mathcal{O}(mb^2 \log b)$ in the worst case. Considering that a sample may contain multiple missing target views, cross-view relation transfer, reliability scoring, and candidate selection require $\mathcal{O}\big(bm^2 k \log(mk)\big)$, where k is the number of selected neighbors. The nonlinear refinement networks and view-specific encoders employ fully connected layers, resulting in $\mathcal{O}\big(b \sum_{v=1}^{m} D_v^2\big)$ complexity. During completion pre-training, calculating $\mathcal{L}_{\text{comp}}$ requires $\mathcal{O}(bmkd_{\max})$, which is dominated by pairwise distance construction when $k \leq b$ and therefore does not change the asymptotic complexity of the completion stage.

For high-order semantic structure mining, constructing the view-wise affinity matrices requires $\mathcal{O}(mb^2 d_e)$, where d_e is the embedding dimension. Computing the t-step transition matrices through dense matrix exponentiation requires $\mathcal{O}(mb^3 \log t)$, where t is the random-walk step number. The structural constraint calculates similarity matrices for all ordered view pairs and requires $\mathcal{O}(m^2 b^2 d_e)$. The label-semantic-consistency-based quality assessment requires $\mathcal{O}\big(b^2\ell + mb^2 d_e + bmd_e^2\big)$, where ℓ is the number of labels.

After merging terms of the same order and omitting dominated lower-order operations, the overall time complexity per main-training iteration is: $\mathcal{O}\big(mb^3 \log t + b^2\big[m(d_{\max} + \log b) + m^2 d_e + \ell\big] + b\big[m^2 k \log(mk) + \sum_{v=1}^{m} D_v^2 + md_e^2\big]\big)$. The cubic term results from dense matrix exponentiation and is the highest-order operation with respect to the batch size. During the warm-up stage, this term is omitted because the high-order structure matrices are initialized as identity matrices.

The space complexity is mainly determined by model parameters, sample-relation matrices, and intermediate feature representations. The model parameters require $\mathcal{O}(N_\theta^{\text{all}})$ storage, where N_θ^{all} includes both trainable and frozen parameters. The batch-level distance, transition, high-order structure, label-similarity, and cross-view similarity matrices require at most $\mathcal{O}(m^2 b^2)$ space. The intermediate matrices involved in dense matrix exponentiation require $\mathcal{O}(mb^2 \log t)$, while the temporary pairwise difference tensor used for distance calculation requires $\mathcal{O}(b^2 d_{\max})$. The cached neighbor features, completed features, view-specific embeddings, and predictions require $\mathcal{O}\big(b[mkd_{\max} + md_{\max} + md_e + \ell]\big)$.

After combining terms of the same order, the overall space complexity is: $\mathcal{O}\big(N_\theta^{\text{all}} + b^2\big[m^2 + m \log t + d_{\max}\big] + b\big[m(k+1)d_{\max} + md_e + \ell\big]\big)$. Thus, the principal memory cost grows quadratically with the batch size, while the dependence on the total number of samples is controlled through mini-batch processing.

To assess practical efficiency, we compare ReCHSM with DICNet, RANK, and DSPNF on Corel5k and Mirflickr under the same platform, data partitions, missing masks, batch size,

Table 8 Time cost, peak GPU memory, and parameter analysis on Corel5k and Mirflickr under 50% missing views, 50% missing labels, and 70% training samples

Dataset	Method / Measure	DICNet	DSPNF	RANK	ReCHSM
Corel5k	Training (s)	511.70	520.92	540.62	1799.07
	Inference (s)	0.17	0.30	0.27	1.28
	Peak GPU memory (MB)	2208.00	1822.00	3992.00	1374.00
	Parameters (M)	111.14	99.05	191.78	75.96
	Completion (s)	–	–	–	1242.14
	Random walk (s)	–	–	–	10.14
Mirflickr	Training (s)	1090.49	924.49	2787.07	7960.01
	Inference (s)	0.43	0.61	0.70	7.33
	Peak GPU memory (MB)	2626.00	2182.00	16738.00	1574.00
	Parameters (M)	131.12	116.95	191.66	86.72
	Completion (s)	–	–	–	6522.44
	Random walk (s)	–	–	–	55.04

and main training epochs. As shown in Table 8, ReCHSM requires more training and inference time because it includes 50 completion pre-training epochs. However, it uses the fewest parameters and the lowest peak GPU memory on both datasets while achieving superior classification performance.

5 Conclusion

This paper proposes ReCHSM, a reliable cross-view neighborhood relation transfer and high-order semantic structure mining framework for IMvPMLC. This framework combines three key components. First, we develop a completion mechanism based on reliable cross-view neighborhood relation transfer. This mechanism transfers reliable instance relations between existing views to the missing view through a reliable neighborhood scoring mechanism, and further refines the initial reconstructed data using a lightweight MLP nonlinear refinement network, thus obtaining a robust reconstructed view. Furthermore, we introduce a high-order semantic structure mining strategy based on random walks, propagating the repaired local relations to the more informative global topological dependencies to guide the encoder in extracting more compact and discriminative high-level semantic features. Finally, considering that the learned semantic information should serve the final label classification, we design a view quality assessment module based on label semantic consistency to learn sample-level adaptive fusion weights to effectively aggregate multi-view information. Extensive results show that ReCHSM consistently outperforms or remains competitive with existing approaches under dual-incomplete, single-incomplete, and complete-data settings. Further ablation experiments validate the indispensability of individual components, particularly underscoring the strong complementarity between reliable view completion and high-order semantic structure modeling.

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Wenbin Bu: Visualization, Validation, Experimental assistance, Writing - review & editing.
Nisuo Du: Supervision, Funding acquisition, Writing - review & editing.
Xiaohuan Lu: Supervision, Writing - review & editing.
All authors read and approved the final manuscript.

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Data Availability The datasets used in this study are publicly available benchmark datasets. The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethical Approval Not applicable.

Competing Interests The authors declare no competing interests.

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